MINJAE KWON

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Research Interests

Engineering nanostructures that shape light and fluidic behavior, guided by computational modeling.

Nanofabrication: periodic/3D nanostructures with tailored optical, mechanical, and interfacial properties;
Plasmonics & Metasurfaces: engineered for plasmonic field enhancement and optical sensing.
Electrokinetics: structure-mediated particle and fluid control via electrokinetic (DEP/ACEO) phenomena;
FEA Simulation: finite element analysis for optical and electrical field modeling.

Education

Korea University (Seoul, Republic of Korea) — Sep. 2025 – Present
Master's Student in Department of Bioengineering
- Advised by Prof. Yong-Sang Ryu

Korea University (Seoul, Republic of Korea) — Mar. 2023 – Aug. 2025
B.S. in Biomedical Engineering & Electrical Engineering (Double Major)
- GPA : N.A / 4.50

Konkuk University (Seoul, Republic of Korea) — Mar. 2017 – Feb. 2023
Majored in Electronic and Electrical Engineering
- Transferred to Korea University
- Leave of absence due to national military service: Jul. 2019 – Jan. 2021

Publications

[1] S. Lee‡, S. Choi‡, T. Kim, **M. Kwon**, T. J. Mun, S. Son, H. Seong, K. Lee, H. Im, E.-S. Yu, M.-K. Kim*, and Y.-S. Ryu*, *"Analyte-accessible field volume as a governing design principle for nanophotonic sensing,"* under review, Advanced Materials (2026).

‡Equal contribution, *Corresponding author

Research Experience

Ryu group, Korea University (Seoul, Republic of Korea) — Aug. 2024 – Present
Undergraduate/Graduate Student Researcher *(Advised by Prof. Yong-Sang Ryu)*
- **Current work**: Designing and fabricating periodic MIM nanohole plasmonic metasurfaces with geometric control — nanohole period and diameter defined by lithography, and insulator and metal layer thicknesses controlled to within 5 nm by ALD and electron-beam evaporation — and simulating optical/electrokinetic fields to identify geometries yielding up to ~1.8×10^5-fold field enhancement at sensing hotspots. Building on the prior co-authored work, engineering the chip to isolate one nanoparticle per hotspot via DEP/ACEO transport for single-particle SERS detection, and applying CNNs to SEM images to quantify capture uniformity (RSD) as a measure of performance.
- **Collaborative projects**: Leading full-chip fabrication of an electrochemical-sensing MIM nanostructure and providing nanofabrication support for concurrent lab projects (a PRET-based metasurface, a hydrogen-sensing MEMS platform); and, as a Collaborating Researcher at the Nanophotonics Research Center, Korea Institute of Science and Technology (KIST), contributing to a KU-KIST joint project developing label-free plasmonic sensing platforms for GPCR conformational dynamics in a drug-discovery initiative.

Radicheck (Seoul, Republic of Korea) Mar. 2024 – Aug. 2025
<u>Founder & Chief Technology Officer</u>

- Developed an integrated radiation detection platform featuring real-time data-driven labeling systems for seafood safety assessment; two patents are currently pending on this platform.

Technical Proficiency

Fabrication & Characterization Skills

- <u>Lithography</u>: Aligner-assisted Photolithography, Laser Interference Lithography (LIL), Electron Beam Lithography (EBL), Soft Lithography, Nanoimprint Lithography (NIL)
- <u>PVD (Physical Vapor Deposition)</u>: Thermal Deposition, Electron Beam Deposition, Sputter Deposition
- <u>CVD (Chemical Vapor Deposition)</u>: Atomic Layer Deposition (ALD), Plasma Enhanced Chemical Vapor Deposition (PECVD)
- <u>Etching</u>: Reactive Ion Etching (RIE), Inductively Coupled Plasma Reactive Ion Etching (ICP-RIE)
- <u>Surface Treatment</u>: Oxygen Plasma Ashing, UV-Ozone
- <u>Material / Structure Analysis</u>: Scanning Electron Microscope (SEM), Atomic Force Microscope (AFM), Energy Dispersive X-ray Spectroscopy (EDS), Stylus / Optical Surface Profiler, Spectroscopic Ellipsometry, Reflectance Spectroscopy, Raman Spectroscopy, Biological / Metallographic Microscope

Computational & Design Skills

- <u>Language:</u> Python, C/C++, LaTeX, MATLAB
- <u>FEA Simulation:</u> COMSOL Multiphysics, Lumerical FDTD
- <u>Data Analysis:</u> OriginLab, Veusz, Python-based data analysis (PyTorch, OpenCV)
- <u>2D/3D Design:</u> Rhino, AutoCAD, Keyshot
- <u>Physical Design</u>: PCB/Breadboard Circuit Design, 3D Printing, Metal / Polymer CNC, Metal / Wafer Laser Cutting

Leadership & Extracurricular Activities (Selected)

School of Biomedical Engineering, Korea University

<u>Founder & President</u>, KU Biomedical Eng. Abroad Mentorship (KUBEAM) Aug. 2025 – Present

- Founded an initiative connecting 80+ biomedical engineering students and alumni to pursue opportunities abroad, such as exchange programs, Ph.D. applications, and international careers.
- Built a curated study-abroad resource hub, alumni interviews, and member database — establishing KUBEAM as a de facto network connecting overseas alumni and faculty with current applicants.

<u>Biosensor Team Lead Organizer</u>, Biomedical Engineering Academic Team (BEAT) Aug. 2025 – Jul. 2026

- Led and mentored 60+ undergraduates in acquiring the concepts and technical skills necessary to explore the field of biosensors (e.g., SEM imaging, data analysis, critical literature review, and journal-club presentations).

<u>Senior Advisor & Vice President</u>, Department Student Council Nov. 2024 – Nov. 2025

- Advised and supported planning and execution of academic, cultural, and community events for 500+ undergraduates.
- Led a 30+ member committee to organize the department alumni day, securing over $22,000 in departmental and corporate sponsorships to deliver a large-scale event.

Creative Challenger Program, Korea University Aug. 2023 – Feb. 2024

<u>Project Leader</u>, Vine Robot Colonoscopy Project (Advised by Prof. Daehee Hong and Dr. Sang-Hyun Kim)

- Developed and validated a novel soft-robot colonoscopy platform; conducted in-vivo experiments in collaboration with the Department of Gastroenterology, Korea University Medical Center.
- Recognized with Excellence Awards at the 15th Creative Challenger Program and the 2nd Changemakers Forum for the platform.

Office of International Affairs, Konkuk University Jul. 2021 – Aug. 2022

<u>President</u>, International Student Volunteer (ISV)

- Assisted inbound and outbound exchange students by providing language support, administrative assistance, and organizing cultural exchange programs. Accommodated and supported over 600 students throughout the term.

65th Medical Brigade, 8th U.S. Army Jul. 2019 – Jan. 2021

<u>Combat Medic & Medical Support Assistant</u>, BDAACH (549th Hospital Center)

- Selected for the KATUSA (Korean Augmentation to the U.S. Army) program. Served in the Physical Performance Service Line for outpatient care and on the COVID-19 Response Team for quarantine operations and vaccine administration.
- Contributed to the accountability of over $5M worth of medical equipment and supplies for the entire clinic. Honorably discharged as Sergeant (E-5); recipient of the Army Achievement Medal and Army Commendation Medal.

Awards and Scholarships

BK21 FOUR Graduate Research Scholarship, Ministry of Education (Full Tuition)	Sep. 2025 – Present
Best Poster Presentation Award, Korean Biochip Fall Symposium	Nov. 2025
Outstanding New Student Scholarships, Korea University	Sep. 2025
Dean's List (Highest Honors, 2 semesters), Korea University	Mar. 2023 – Jun. 2025
Excellence Award, 2nd Changemakers Forum	Nov. 2024
Best Startup Award, Korea University	Nov. 2024
Third Place, Capstone Design Fair	Sep. 2024
Excellence Award, 15th Creative Challenger Program	Feb. 2024

Languages

Korean	Native speaker
English	TOEFL 5.0 / 6.0 (equivalent to 100)

Teaching Experience

Teaching Assistant — Prof. Yong-Sang Ryu
BMED442, Biomedical Semiconductor Process — Fall 2025, Fall 2026

Teaching Assistant — Prof. Yong-Sang Ryu
HBL514, Biomedical Design — Spring 2026